Water Scarcity and Pollution

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Learning Objectives

- After taking this class, the students will be able to:
  - *acknowledge* the water resources problems we are facing today
  - *identify* different types of water pollution and *articulate* why they matters
  - *acknowledge* the challenges in dealing with the water problems

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Water Problem I: Water Scarcity

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1. Facts about Water Scarcity in China and the World

1.1 Diverse manifestations and hidden nature of water scarcity





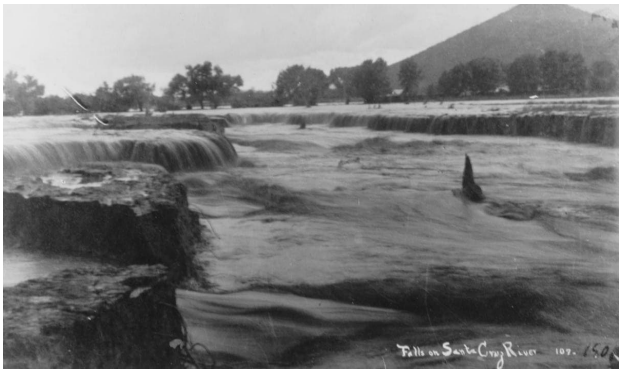
Tucson, Arizona

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1. Facts about Water Scarcity in China and the World

1.1 Diverse manifestations and hidden nature of water scarcity





Santa Cruz River, Tucson, Arizona

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Los Angeles Times

CALIFORNIA

Unprecedented water restrictions hit Southern California today: What they mean to you



A barren median on North Citrus Avenue in Covina. More than 97% of the state is now under severe, extreme or exceptional drought (Robert Gauthier / Los Angeles Times)

By Hayley Smith
Staff Writer

June 1, 2022 Updated 6:38 AM PT

- The MWD board voted unanimously to adopt the emergency measures to “reduce non-essential water use” in certain areas.
- outdoor watering accounts for roughly half of all urban water use.
- restricting outdoor watering: nearly everyone in the city of L.A. — to two-day-a-week watering at only 8 minutes per station per day, or two 15 minute-cycles per watering day for sprinklers with water-conserving nozzles.

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1. Facts about Water Scarcity in China and the World

1.1 Diverse manifestations and hidden nature of water scarcity





Mount Tanglang, Shenzhen

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1. Facts about Water Scarcity in China and the World

1.1 Diverse manifestations and hidden nature of water scarcity

Let water saving be the public etiquette of Shenzhen citizens



Shenzhen faces historical drought with daily water shortage of 1 million cubic meters

By Global Times

Published: Dec 10, 2021 12:34 AM

让节约用水成为深圳人的文明素养

深圳向全体市民发出节水倡议

来源：深圳特区报　2021年12月08日　版次：A02

深圳特区报讯（记者 方胜）深圳境内没有大江大河，全市90%以上的饮用水从东江引入。去年和今年连续两年干旱，深圳用水量却明显增长，今冬明春供水安全形势不容乐观。

为提高水资源节约集约利用效率，保障水安全，12月7日，深圳市节约用水领导小组办公室发出倡议，号召全体市民和企事业单位增强节水意识，养成节水习惯，使用节水器具，加大再生水的使用，加大节水宣传。倡议书全文如下。

全体市民朋友，各企事业单位：

深圳地处华南沿海，境内没有大江大河，本地水资源短缺，全市90%以上用水依赖东江引入。由于去年和今年连续两年干旱，东江流域正遭受1963年以来最严峻旱情，11月17日，市三防指挥部启动水利抗旱Ⅳ级应急响应，今冬明春供水安全形势不容乐观，全民节约用水刻不容缓。

- Rain-source urban river in Shenzhen
- 169 reservoirs in total： 18 municipal reservoirs; 151 district-managed reservoirs



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1. Facts about Water Scarcity in China and the World

1.1 Diverse manifestations and hidden nature of water scarcity



Hai River, Tianjin, China

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Reflective Question: What have you learned from the above information?

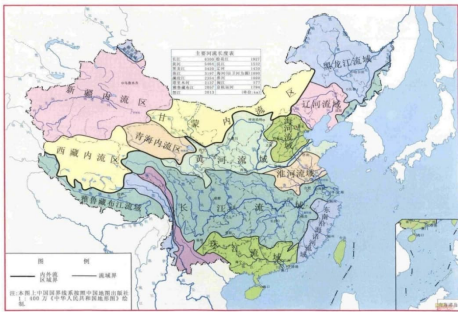
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1. Facts about Water Scarcity in China and the World

1.2 The standards and indicators of water scarcity

Make a guess:
Which province is the thirstiest in our country?

- □ Gansu Province
- □ Shaanxi Province
- □ Henan Province
- □ Ningxia Province
- □ Hebei Province
- □ Shandong Province



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Water Resources Distribution across China

Province	Water Resources per capita (m3)	Total Volume of Water Resources (billion m3)	Population	RDP
Tianjin	51.86	8.1	1562	14104.28
Beijing	114.21	24.6	2154	35371.28
Hebei	149.50	113.5	7592	35104.52
Henan	174.90	168.6	9640	54259.2
Ningxia	181.30	12.6	695	3748.48
Shandong	193.84	195.2	10070	71067.53
Shanghai	198.93	48.3	2428	38155.32
Shanxi	260.93	97.3	3729	17026.68
Jiangsu	287.11	231.7	8070	99631.52
Liaoning	588.24	256	4352	24909.45
Anhui	848.10	539.9	6366	37113.98
Hubei	1035.43	613.7	5927	45828.31
Gansu	1231.21	325.9	2647	8718.3
Shannxi	1277.86	495.3	3876	25793.17
Chongqing	1594.43	498.1	3124	23605.77

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Water Resources Distribution across China

Province	Water Resources per capita (m3)	Total Volume of Water Resources (billion m3)	Population	RDP
Inner Mongolia	1763.39	447.9	2540	17212.53
Guangdong	1795.16	2068.2	11521	107671.07
Jilin	1880.71	506.1	2691	11726.82
Zhejiang	2258.97	1321.5	5850	62351.74
Hainan	2669.84	252.3	945	5308.93
Hunan	3033.10	2098.3	6918	39752.12
Guizhou	3083.08	1117	3623	16769.34
Yunnan	3157.27	1533.8	4858	23223.75
Sichuan	3282.27	2748.9	8375	46615.82
Fujian	3432.92	1363.9	3973	42395
Xinjiang	3448.67	870.1	2523	13597.11
Heilongjiang	4029.33	1511.4	3751	13612.68
Guangxi	4244.15	2105.1	4960	21237.14
Jiangxi	4396.91	2051.6	4666	24747.5
Qinghai	15120.07	919.3	608	2965.95
Tibet	128116.8	4496.9	351	1697.82

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1. Facts about Water Scarcity in China and the World

1.2 The standards and indicators of water scarcity

Average water resources per capita (China): 1998 $m^3/person$ (2019)

Average water resources per capita (World): 5555 $m^3/person$ (2019)

According to the UN standard,

- Water Stress: An area is experiencing water stress when annual water supplies drop below 1700 m^3 per person.
- Water Scarcity: When annual water supplies drop below 1000 m^3 per person, the population faces water scarcity.
- Absolute Scarcity: When annual water supplies drop below 500 m^3 per person, the population faces "absolute scarcity".

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1. Facts about Water Scarcity in China and the World

1.2 The standards and indicators of water scarcity

Do you know how much water we consume each day?





https://thewaterweeat.com/

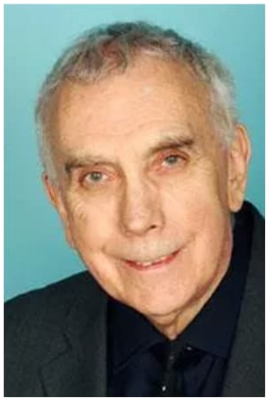
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1. The Facts about Water Scarcity in China and the World

1.2 The standards and indicators of water scarcity

Water Footprint （水足迹） and Virtual Water（虚拟水）

- What is water footprint? An Assessment of Water Use Patterns at Different Levels, Both *Direct* and *Virtual*.
- It takes water – a lot of it – to produce food, to make energy and to manufacture consumer products. This is what's known as *virtual water*.
- What is your water footprint? Play with this water footprint calculator.



Prof. Tony Allan (1937-2021)

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Takeaway

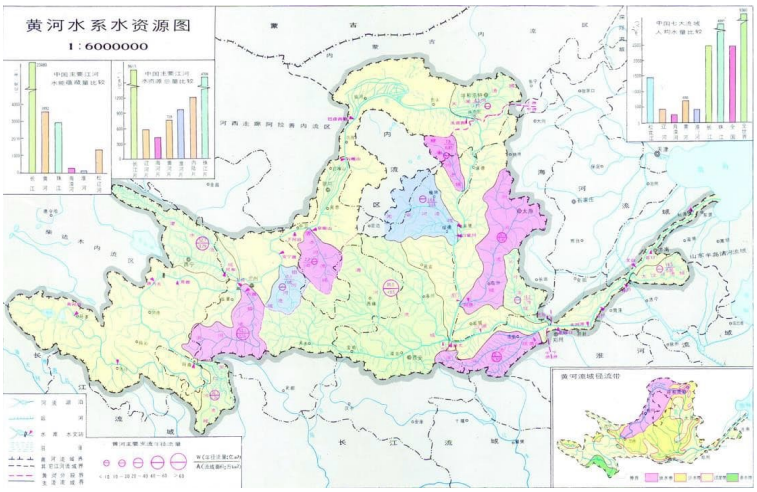
- Water scarcity has diverse manifestations and can be very deceptive.
- According to the UN standard, nearly half of the provinces in our country are under water stress; about 1/3 of the provinces face water scarcity issues, and 1/4 are at the “absolute scarcity” level.
- Water consumption can be both direct and virtual. We consume more water than we usually think.

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If water resource is so scarce, what could we do?

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Yellow River Basin and Water Scarcity



- The Yellow River only share 2.5% of the total water resources in China, while feeds 12% of population, 15% arable lands, and more than 50 municipalities in China.
- It runs across 9 provinces and supports the water use of 11 province-level unit.
- Rapid population growth and economic development along river intensify the water demands.
- In 1995-1998, the Yellow River failed to reach the sea （断流） for about 120 days each year, in some years failing even to reach Shandong Province.

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Water Use Regulation on Yellow River

Province	Water Supply Amount (100million cube meter)
Qinghai	14.1
Sichuan	0.4
Gansu	30.4
Ningxia	40
Inner Mongolia	58.6
Shannxi	38
Shanxi	43.1
Henan	55.4
Shandong	70
Hebei	20
Tianjin	

Water Allocation Program of the Yellow River （1987）

- The MWR coordinated and issued the “Water Allocation Program of the Yellow River”(《黄河可供水量分配方案》), which was approved by the State Council in 1987. This water allocation program mainly considered the actual amount of water used by various users in 1980 and the expected water demand in 2000 for each province respectively.
- In 1998 the YRCC promulgated more strict provincial water use regulations and began planning a sophisticated system to monitor water use and flow conditions in near-real time. [《黄河可供水量年度分配及干流水量调度方案》《黄河水量调度管理办法》]
- Yellow River Commission of Conservancy (YRCC, 黄河水利委员会) negotiate with the department of the water resources of the 11 provinces to determine the allocation of water each year.
- These efforts have been successful in halting the desiccation of the Yellow River, which has continually reached the sea since 1999, though flows remain below levels considered necessary to satisfy ecological needs.

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Problematic Implementation and Water Rights Disputes

表10 2020年黄河供水区省（区）分行业地表水取、耗水量

单位：亿立方米

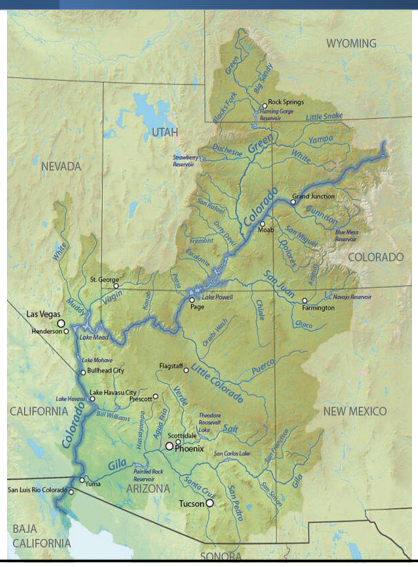
省(区)	项 目	合 计	农 业	工 业	生 活	生态环境
青 海	取水量	12.64	9.80	0.18	1.36	1.30
	耗水量	9.33	6.98	0.11	1.01	1.23
四 川	取水量	0.24	0.15	0.01	0.08	0.00
	耗水量	0.19	0.12	0.01	0.06	0.00
甘 肃	取水量	34.52	23.68	2.87	4.56	3.41
	耗水量	28.03	19.17	2.18	3.33	3.35
宁 夏	取水量	65.07	56.18	3.80	1.74	3.35
	耗水量	39.87	32.49	3.13	1.23	3.02
内蒙古	取水量	88.31	73.50	4.76	2.34	7.71
	耗水量	66.15	55.70	4.18	1.86	4.41
陕 西	取水量	36.56	19.06	4.73	8.48	4.29
	耗水量	29.22	16.24	3.11	5.58	4.29
山 西	取水量	32.15	18.16	4.60	4.38	5.01
	耗水量	28.99	16.20	4.14	3.64	5.01
河 南	取水量	52.67	32.17	6.25	4.19	10.06
	耗水量	49.76	30.87	5.10	3.92	9.87
山 东	取水量	85.68	44.90	14.32	15.75	10.71
	耗水量	83.96	44.00	13.91	15.49	10.56
河 北	取水量	18.33	9.24	/	3.05	6.04
	耗水量	18.33	9.24	/	3.05	6.04
合 计	取水量	426.17	286.84	41.52	45.93	51.88
	耗水量	353.83	231.01	35.87	39.17	47.78

- Some localities do not put into practice the water quantity allocation and dispatch plan and exceed the allocation limits. Many localities extract water from tributaries of the Yellow before it enters the mainstream. (YRCC, 2011)
- Industrial water quotas had almost reached their limits and the remaining quotas were not sufficient for new industrial projects, especially in energy bases.

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Colorado River and Interstate Water Wars

- The Colorado River is the seventh largest river in the U.S., supplying water to over 30 million people.
  - Upper river: Wyoming, Colorado, Utah, and New Mexico
  - Lower river: Arizona, Nevada, California
- The 1922 *Colorado River Compact* allocated 7.5 million acre-feet (one acre-foot is roughly 325,000 gallons) to California, Nevada and Arizona, and another 7.5 million acre-feet to Utah, Wyoming, Colorado and New Mexico.



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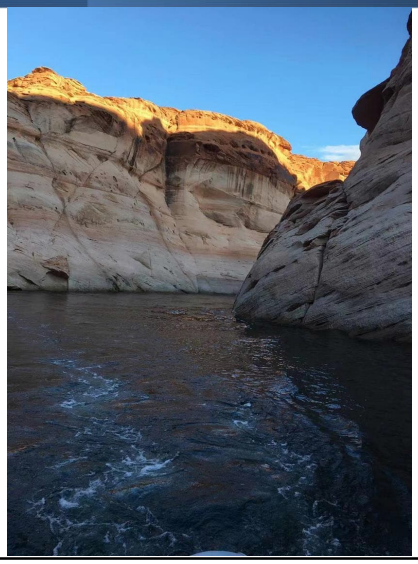
Colorado River and Interstate Water Wars

However, estimates based on tree ring analysis have determined that the actual yearly flow of the river over the last 1,200 years is roughly 14.6 million acre-feet.

Climate stresses are raising the stakes. Rising temperatures require farmers to use more water to grow the same amount of crops. Prolonged and severe droughts decrease available supplies.

The "Law of the River" – the body of rules, regulations and laws governing the Colorado River – has allocated more water to the states than the river reliably provides.

The Upper Basin states want to use more water, which also accelerates the disputes among the states.



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Water Problem II: Water Pollution

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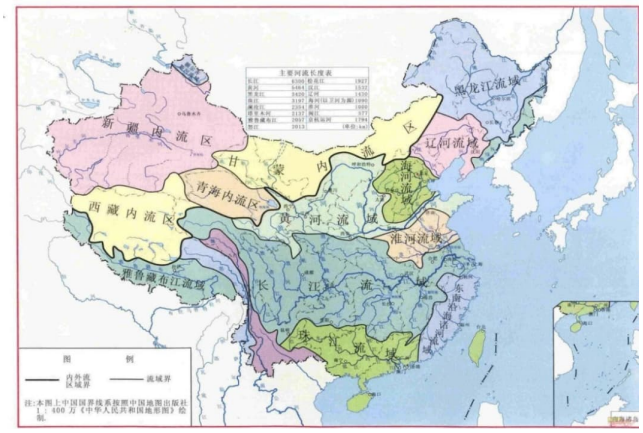
What is water pollution?

- Water pollution occurs when harmful substances—often chemicals or microorganisms—contaminate a stream, river, lake, ocean, aquifer, or other body of water, degrading water quality and rendering it toxic to humans or the environment.

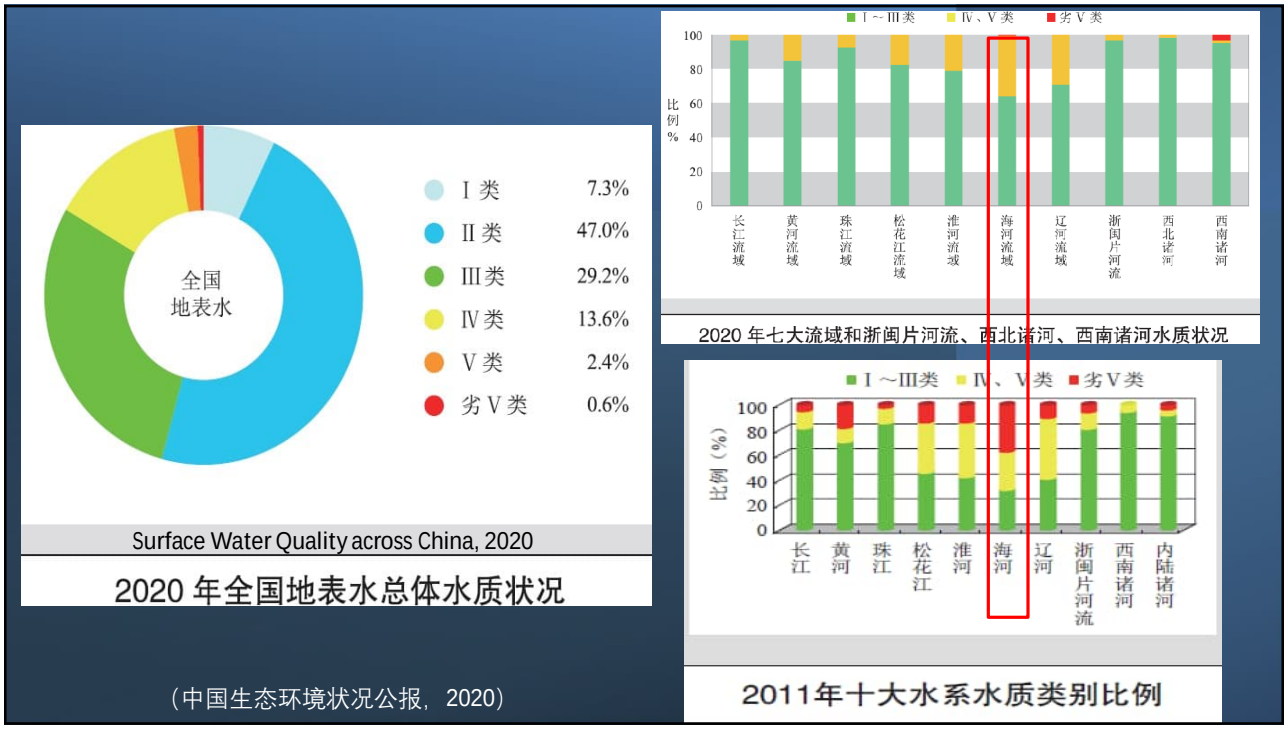
- Types of water pollution:
  - Point source pollution
  - Nonpoint source pollution

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Pop-up Question 2: Can you guess which river basin is most polluted?



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Point Source Pollution (固定污染源污染）

- Point pollution: Any single identifiable source of pollution from which pollutants are discharged, such as a pipe, ditch, ship or factory smokestack.
- Factories and sewage treatment plants are two common types of point sources.
  - Factories typically discharge one or more pollutants in their discharged waters (called effluents).
  - Sewage treatment plants treat human wastes and send the treated effluent to a stream or river.

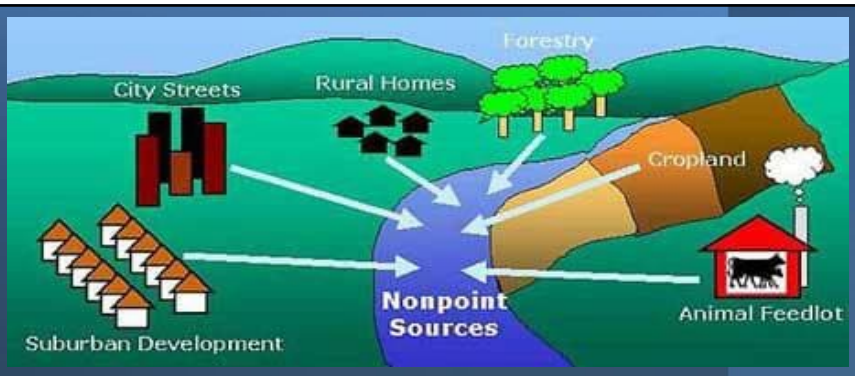


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Nonpoint Source Pollution （面源污染）

- Nonpoint source pollution is pollution in the water that does not come from one specific place.
- It can be water that falls on the ground and moves across the land, picking up pollutants along the way. This water is called runoff. It may eventually end up in a river, a lake, or the ocean.
- Although the concentration of some pollutants from runoff may be lower than the concentration from a point source, the total amount of a pollutant delivered from nonpoint sources may be higher because the pollutants come from many places.

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Pop-up Question 3: Can you guess which type of water pollution is more difficult to deal with?

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Pollutants	Year	Industrial pollution	Agricultural pollution	Urban domestic pollution	centralized pollution control and treatment facilities
COD (10,000 ton)	2011	354.8	1186.1	938.8	20.1
	2012	338.5	1153.8	912.8	18.7
	2013	319.5	1125.8	889.8	17.7
	2014	311.3	1102.4	864.4	16.5
	2015	293.5	1068.6	846.9	14.5
	2016	122.8	57.1	473.5	4.6
	2017	91	31.8	483.8	2.3
	2018	81.4	24.5	476.8	1.5
	2019	77.2	18.6	469.9	1.4

(Annual Report of Environmental Statistics, 2015; Annual Report of Ecology and Environmental Statistics 2016-2019)

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Pollutants	Year	Industrial pollution	Agricultural pollution	Urban domestic pollution	centralized pollution control and treatment facilities
NH-3	2011	28.1	82.7	147.7	2
	2012	26.4	80.6	144.6	1.9
	2013	24.6	77.9	141.4	1.8
	2014	23.2	75.5	138.1	1.7
	2015	21.7	72.6	134.1	1.5
	2016	6.5	1.3	48.4	0.7
	2017	4.4	0.7	45.4	0.3
	2018	4	0.5	44.7	0.2
	2019	3.5	0.4	42.1	0.3

(Annual Report of Environmental Statistics, 2015; Annual Report of Ecology and Environmental Statistics 2016-2019)

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Groundwater Pollution (地下水污染)

- Following the survey of 2013 underground wells by the Ministry of Water Resources, more than 80% of the underground water in large river basins of mainland China is unfit for drinking or bathing because of contamination from industry and farming.

表 8　各省区地下水水质评价结果统计表

行政区	测站总数	2015 年全年			2015 年枯水期	2015 年丰水期	2014 年全年
		Ⅰ至Ⅲ类%	Ⅳ类%	Ⅴ类%	Ⅰ至Ⅲ类%	Ⅰ至Ⅲ类%	Ⅰ至Ⅲ类%
北京	124	53.2	23.4	23.4	54	75.4	26.4
天津	63	14.3	30.2	55.6	11.1	15.9	10.7
河北	203	14.3	36.5	49.3	12.3	31	8.1
山西	190	17.4	34.2	48.4	15.8	22.6	17.9
内蒙古	94	4.3	31.9	63.8	2.1	6.4	0
辽宁	126	7.1	24.6	68.3	6.3	4.8	2.4
吉林	170	18.2	33.5	48.2	20	18.8	16.5
黑龙江	98	9.3	28.9	61.9	16.3	11.3	14.3
江苏	40	32.5	40	27.5	33.3	42.5	27.5
安徽	70	15.5	66.2	18.3	11.4	15.7	12
山东	325	23.1	27.1	49.8	23.1	22.9	19.6
河南	222	7.7	47.7	44.6	9	9.5	7.7
湖北	11	9.1	27.3	63.6	9.1	0	0
陕西	185	33.5	30.3	36.2	35.7	33	34.6
甘肃	116	25.9	22.4	51.7	20.7	19	16.4
青海	15	73.3	6.7	20	71.4	64.3	53.3
宁夏	33	0	33.3	66.7	0	0	0
新疆	18	44.4	22.2	33.3	33.3	55.6	5.6

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Transboundary Water Pollution（跨界水污染）

- Transboundary pollution is defined legally as pollution that originates in one jurisdiction but can cause damage in another jurisdiction's environment, by crossing borders through water pathways.
- A distinctive characteristic of transboundary pollution problems is that pollution does not remain within political boundaries. Thus, this fluid nature of the environment has posed a challenge for environmental governance within this system.

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Polluting thy Neighbors

- In the US, cancer rates are substantially higher in border counties than in interior ones due to transboundary pollution. (Kahn, 2004)
- In Hebei province, polluting firms are more likely to set up in border counties than in interior ones. (Duviver and Xiong, 2013)
- Most downstream county of a province has up to 20 percent more water polluting activities than otherwise identical counties since 2001. (Cai, Chen, Gong, 2016)

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Class Activity:
Can you illustrate why transboundary water pollution is likely to occur?

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Transboundary Pollution and Negative Externalities

- Recap: negative externality
  - Pollution is termed an externality because it imposes costs on people who are "external" to the producer and consumer of the polluting product.
- When water pollution generated by the upstream localities flow through the waterway without sufficient treatment, it causes negative externalities to the downstream localities.
- People located at regional borders may disproportionately suffer from pollution.

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